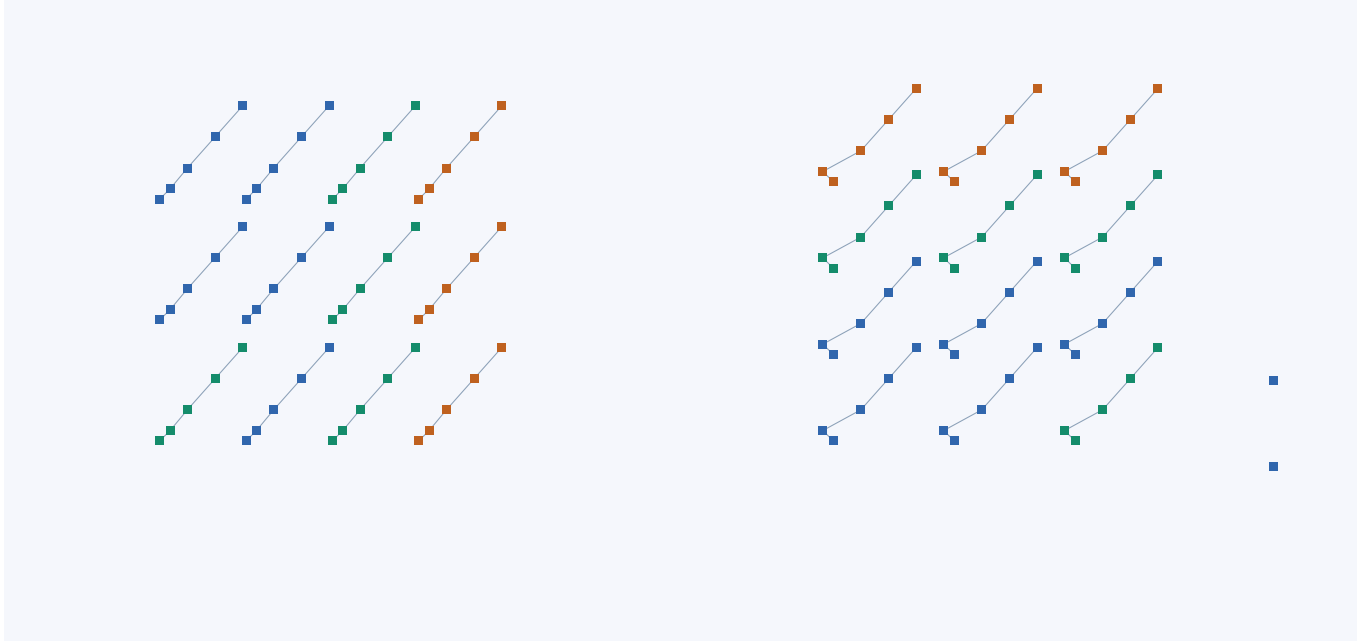


Dual-chiplet package / synthetic demonstrator

Status: PLANNING CHECKS PASS - NOT SIGNOFF

122 sites | 96 assignments | 25,920 um total L1 | 0 crossings

0 errors | 0 warnings | Score 25,920



Physical XY projection; assignments are planning lines, not routed conductors.

Coordinate convention

Canonical unit: micrometre (um). Positive Y is up. Die ports use local coordinates: mirror local X, rotate counter-clockwise around the local origin, then translate to package XY.

Display separation never affects scoring.

Review boundary

Not signoff. Manhattan length and straight-line crossings are proxies, not physical routing or electrical simulation. Ground proximity and power quotas do not certify SI, PI, EM, thermal, or manufacturability. The local review ID is not a cryptographic signature.

Mechanical interfaces

CORE: origin (200, 300) um; size 2300 x 2500 um; R0; mirror X false

IO: origin (6200, 300) um; size 2300 x 2500 um; R90; mirror X false

Interface assignments

CORE:pad:1 -> CORE:bump:1

Net GND | GND | ground | L1 120 um

CORE:bump:1 -> CORE:interposer:1

Net GND | GND | ground | L1 220 um

CORE:ball:1 -> CORE:pcb:1

Net GND | GND | ground | L1 340 um

CORE:pad:2 -> CORE:bump:2

Net CORE_CLK | CORE | clock | L1 120 um

CORE:bump:2 -> CORE:interposer:2

Net CORE_CLK | CORE | clock | L1 220 um

CORE:ball:2 -> CORE:pcb:2

Net CORE_CLK | CORE | clock | L1 340 um

CORE:pad:3 -> CORE:bump:3

Net GND | GND | ground | L1 120 um

CORE:bump:3 -> CORE:interposer:3

Net GND | GND | ground | L1 220 um

CORE:ball:3 -> CORE:pcb:3

Net GND | GND | ground | L1 340 um

CORE:pad:4 -> CORE:bump:4

Net VDD_CORE | CORE | power | L1 120 um

CORE:bump:4 -> CORE:interposer:4

Net VDD_CORE | CORE | power | L1 220 um

CORE:ball:4 -> CORE:pcb:4

Net VDD_CORE | CORE | power | L1 340 um

CORE:pad:5 -> CORE:bump:5

Net CORE_TX_P | CORE | signal | L1 120 um

CORE:bump:5 -> CORE:interposer:5

Net CORE_TX_P | CORE | signal | L1 220 um

CORE:ball:5 -> CORE:pcb:5

Net CORE_TX_P | CORE | signal | L1 340 um

CORE:pad:6 -> CORE:bump:6

Net CORE_TX_N | CORE | signal | L1 120 um

CORE:bump:6 -> CORE:interposer:6

Net CORE_TX_N | CORE | signal | L1 220 um

CORE:ball:6 -> CORE:pcb:6

Net CORE_TX_N | CORE | signal | L1 340 um

CORE:pad:7 -> CORE:bump:7

Net GND | GND | ground | L1 120 um

CORE:bump:7 -> CORE:interposer:7

Net GND | GND | ground | L1 220 um

CORE:ball:7 -> CORE:pcb:7

Net GND | GND | ground | L1 340 um

CORE:pad:8 -> CORE:bump:8

Net VDD_CORE | CORE | power | L1 120 um

CORE:bump:8 -> CORE:interposer:8

Net VDD_CORE | CORE | power | L1 220 um

CORE:ball:8 -> CORE:pcb:8

Net VDD_CORE | CORE | power | L1 340 um

CORE:pad:9 -> CORE:bump:9

Net CORE_DATA0 | CORE | signal | L1 120 um

CORE:bump:9 -> CORE:interposer:9

Net CORE_DATA0 | CORE | signal | L1 220 um

CORE:ball:9 -> CORE:pcb:9

Net CORE_DATA0 | CORE | signal | L1 340 um

CORE:pad:10 -> CORE:bump:10

Net CORE_DATA1 | CORE | signal | L1 120 um

CORE:bump:10 -> CORE:interposer:10

Net CORE_DATA1 | CORE | signal | L1 220 um

CORE:ball:10 -> CORE:pcb:10

Net CORE_DATA1 | CORE | signal | L1 340 um

CORE:pad:11 -> CORE:bump:11

Net GND | GND | ground | L1 120 um

CORE:bump:11 -> CORE:interposer:11

Net GND | GND | ground | L1 220 um

CORE:ball:11 -> CORE:pcb:11

Net GND | GND | ground | L1 340 um

CORE:pad:12 -> CORE:bump:12

Net VDD_CORE | CORE | power | L1 120 um

CORE:bump:12 -> CORE:interposer:12

Net VDD_CORE | CORE | power | L1 220 um

CORE:ball:12 -> CORE:pcb:12

Net VDD_CORE | CORE | power | L1 340 um

IO:pad:1 -> IO:bump:1

Net GND | GND | ground | L1 120 um

IO:bump:1 -> IO:interposer:1

Net GND | GND | ground | L1 340 um

IO:ball:1 -> IO:pcb:1

Net GND | GND | ground | L1 340 um

IO:pad:2 -> IO:bump:2

Net IO_CLK | IO | clock | L1 120 um

IO:bump:2 -> IO:interposer:2

Net IO_CLK | IO | clock | L1 340 um

IO:ball:2 -> IO:pcb:2

Net IO_CLK | IO | clock | L1 340 um

IO:pad:3 -> IO:bump:3

Net GND | GND | ground | L1 120 um

IO:bump:3 -> IO:interposer:3

Net GND | GND | ground | L1 340 um

IO:ball:3 -> IO:pcb:3

Net GND | GND | ground | L1 340 um

IO:pad:4 -> IO:bump:4

Net VDD_IO | IO | power | L1 120 um

IO:bump:4 -> IO:interposer:4

Net VDD_IO | IO | power | L1 340 um

IO:ball:4 -> IO:pcb:4

Net VDD_IO | IO | power | L1 340 um

IO:pad:5 -> IO:bump:5

Net IO_TX_P | IO | signal | L1 120 um

IO:bump:5 -> IO:interposer:5

Net IO_TX_P | IO | signal | L1 340 um

IO:ball:5 -> IO:pcb:5

Net IO_TX_P | IO | signal | L1 340 um

IO:pad:6 -> IO:bump:6

Net IO_TX_N | IO | signal | L1 120 um

IO:bump:6 -> IO:interposer:6

Net IO_TX_N | IO | signal | L1 340 um

IO:ball:6 -> IO:pcb:6

Net IO_TX_N | IO | signal | L1 340 um

IO:pad:7 -> IO:bump:7

Net GND | GND | ground | L1 120 um

IO:bump:7 -> IO:interposer:7

Net GND | GND | ground | L1 340 um

IO:ball:7 -> IO:pcb:7

Net GND | GND | ground | L1 340 um

IO:pad:8 -> IO:bump:8

Net VDD_IO | IO | power | L1 120 um

IO:bump:8 -> IO:interposer:8

Net VDD_IO | IO | power | L1 340 um

IO:ball:8 -> IO:pcb:8

Net VDD_IO | IO | power | L1 340 um

IO:pad:9 -> IO:bump:9

Net IO_DATA0 | IO | signal | L1 120 um

IO:bump:9 -> IO:interposer:9

Net IO_DATA0 | IO | signal | L1 340 um

IO:ball:9 -> IO:pcb:9

Net IO_DATA0 | IO | signal | L1 340 um

IO:pad:10 -> IO:bump:10

Net IO_DATA1 | IO | signal | L1 120 um

IO:bump:10 -> IO:interposer:10

Net IO_DATA1 | IO | signal | L1 340 um

IO:ball:10 -> IO:pcb:10

Net IO_DATA1 | IO | signal | L1 340 um

IO:pad:11 -> IO:bump:11

Net GND | GND | ground | L1 120 um

IO:bump:11 -> IO:interposer:11

Net GND | GND | ground | L1 340 um

IO:ball:11 -> IO:pcb:11

Net GND | GND | ground | L1 340 um

IO:pad:12 -> IO:bump:12

Net VDD_IO | IO | power | L1 120 um

IO:bump:12 -> IO:interposer:12

Net VDD_IO | IO | power | L1 340 um

IO:ball:12 -> IO:pcb:12

Net VDD_IO | IO | power | L1 340 um

CORE:interposer:5 -> CORE:ball:5

Net CORE_TX_P | CORE | signal | L1 340 um

CORE:interposer:6 -> CORE:ball:6

Net CORE_TX_N | CORE | signal | L1 340 um

IO:interposer:5 -> IO:ball:5

Net IO_TX_P | IO | signal | L1 340 um

IO:interposer:6 -> IO:ball:6

Net IO_TX_N | IO | signal | L1 340 um

CORE:interposer:1 -> CORE:ball:1

Net GND | GND | ground | L1 340 um

CORE:interposer:10 -> CORE:ball:10

Net CORE_DATA1 | CORE | signal | L1 340 um

CORE:interposer:11 -> CORE:ball:11

Net GND | GND | ground | L1 340 um

CORE:interposer:12 -> CORE:ball:12

Net VDD_CORE | CORE | power | L1 340 um

CORE:interposer:2 -> CORE:ball:2

Net CORE_CLK | CORE | clock | L1 340 um

CORE:interposer:3 -> CORE:ball:3

Net GND | GND | ground | L1 340 um

CORE:interposer:4 -> CORE:ball:4

Net VDD_CORE | CORE | power | L1 340 um

CORE:interposer:7 -> CORE:ball:7

Net GND | GND | ground | L1 340 um

CORE:interposer:8 -> CORE:ball:8

Net VDD_CORE | CORE | power | L1 340 um

CORE:interposer:9 -> CORE:ball:9

Net CORE_DATA0 | CORE | signal | L1 340 um

IO:interposer:1 -> IO:ball:1

Net GND | GND | ground | L1 340 um

IO:interposer:10 -> IO:ball:10

Net IO_DATA1 | IO | signal | L1 340 um

IO:interposer:11 -> IO:ball:11

Net GND | GND | ground | L1 340 um

IO:interposer:12 -> IO:ball:12

Net VDD_IO | IO | power | L1 340 um

IO:interposer:2 -> IO:ball:2

Net IO CLK | IO | clock | L1 340 um

IO:interposer:3 -> IO:ball:3

Net GND | GND | ground | L1 340 um

IO:interposer:4 -> IO:ball:4

Net VDD IO | IO | power | L1 340 um

IO:interposer:7 -> IO:ball:7

Net GND | GND | ground | L1 340 um

IO:interposer:8 -> IO:ball:8

Net VDD IO | IO | power | L1 340 um

IO:interposer:9 -> IO:ball:9

Net IO DATA0 | IO | signal | L1 340 um

Configured rules

maxLength: 2600

minDomainSpacing: 250

pairMaxDistance: 650

pairMaxSkew: 150

clockShieldRadius: 750

clockGroundMin: 2

groundRadius: 1500

powerRadius: 1700

```
requirePowerForSignals: true
```

minGroundRatio: 0.2

crossingWeight: 150

maxCrossings: 0

geometryBudget: 1000000

```
terminalKind: pcb
```

```
requireCompletePaths: true
```

allowedStagePairs:

```
[["pad", "bump"], ["pad", "ball"], ["bump", "interposer"], ["bump", "ball"], ["interposer", "ball"], ["ball", "pcb"]]
```

Keep-outs and P/G regions

```
{ "dielid": "CORE", "kinds": [ "pad", "bump" ], "id": "core-corner", "x": 0, "y": 0, "width": 120, "height": 120 }
```

```
{ "kind": "ball", "domain": "", "minGround": 8, "minPower": 6, "id": "package-power-grid", "x": 0, "y": 0, "width": 7600, "height": 5000 }
```

Findings

No configured-rule findings.

Review and approval

Engineering owner:

Reviewer: _____ Approval date: _____

PDF text uses printable ASCII transliteration. Use HTML or JSON for full Unicode labels.